SwiMDiff: Scene-Wide Matching Contrastive Learning With Diffusion Constraint for Remote Sensing Image

Jiayuan Tian, Jie Lei, *Member, IEEE*, Jiaqing Zhang, Weiying Xie, *Senior Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*— With recent advancements in aerospace technology, the volume of unlabeled remote sensing image (RSI) data has increased dramatically. Effectively leveraging this data through self-supervised learning (SSL) is vital in the field of remote sensing. However, current methodologies, particularly contrastive learning (CL), a leading SSL method, encounter specific challenges in this domain. First, CL often mistakenly identifies geographically adjacent samples with similar semantic content as negative pairs, leading to confusion during model training. Second, as an instance-level discriminative task, it tends to neglect the essential fine-grained features and complex details inherent in unstructured RSIs. To overcome these obstacles, we introduce SwiMDiff, a novel self-supervised pretraining framework designed for RSIs. SwiMDiff employs a scene-wide matching approach that effectively recalibrates labels to recognize data from the same scene as false negatives. This adjustment makes CL more applicable to the nuances of remote sensing. In addition, SwiMDiff seamlessly integrates CL with a diffusion model. Through the implementation of pixel-level diffusion constraints, we enhance the encoder's ability to capture both the global semantic information and the fine-grained features of the images more comprehensively. Our proposed framework significantly enriches the information available for downstream tasks in remote sensing. Demonstrating exceptional performance in change detection and land-cover classification tasks, SwiMDiff proves its substantial utility and value in the field of remote sensing.**

***Index Terms*— Contrastive learning (CL), diffusion model, false negative sample, remote sensing image (RSI).**

I. Introduction

REMOTE sensing image (RSI) analysis and interpretation hold paramount significance in the domain of computer vision, encompassing a range of distinct tasks such as land-cover classification [1], [2], change detection [3], [4], [5], and object detection [6], [7], [8]. Such analysis facilitates the monitoring of natural phenomena and human activities on Earth's surface, encompassing domains like land-use surveillance [9], disaster prevention [10], precision agriculture [11], and wildfire detection [12]. By capturing both natural occurrences and human-induced activities, RSI plays an indispensable role in applications spanning geographic information systems, agriculture, environmental science, and myriad other fields.

In the past decades, with the surge in aerospace technology, Earth observation satellites generate terabytes of RSIs daily [13]. Despite this abundance, two primary challenges persist as follows. 1) High specialist manpower requirement: The identification and labeling of RSIs necessarily demand professional researchers, resulting in high costs. 2) The presence of noisy labels: The intrinsic complexity of RSIs makes generating flawless labels during large-scale data annotation challenging. This abundance of large but noisy labels is harmful to many tasks [14]. Addressing these issues, the remote sensing community has shifted focus to automatic feature extraction and analysis from unlabeled RSIs [15], [16], [17]. Self-supervised learning (SSL), exploiting the intrinsic structure of data, emerges as a key method to harness the potential of large-scale unlabeled RSIs.

Early SSL methods largely relied on various pretext tasks [18], such as jigsaw puzzles [19], patch localization [20], and image inpainting [21]. These methods exhibit limited generalizability and are far surpassed by contrastive learning (CL). CL enhances representation by drawing similar instances closer and distancing dissimilar ones [22], [23], [24], [25], [26], [27]. It captures feature representations with high discriminability and strong generalizability, standing out among various SSL methods.

However, efficiently applying CL in remote sensing is hindered by two main obstacles. First, as delineated by the fundamental laws of geography [28], data samples with close geographical proximity should inherently exhibit a degree of similarity. As depicted in Fig. 1, images from the same scene demonstrate significant semantic and perceptual similarities. However, the current CL paradigm tends to classify geographically and semantically similar samples as negatives, overlooking their potential mutual connections and resulting in sample confusion [29]. Second, RSIs often lack clear foreground-background distinctions, with key information

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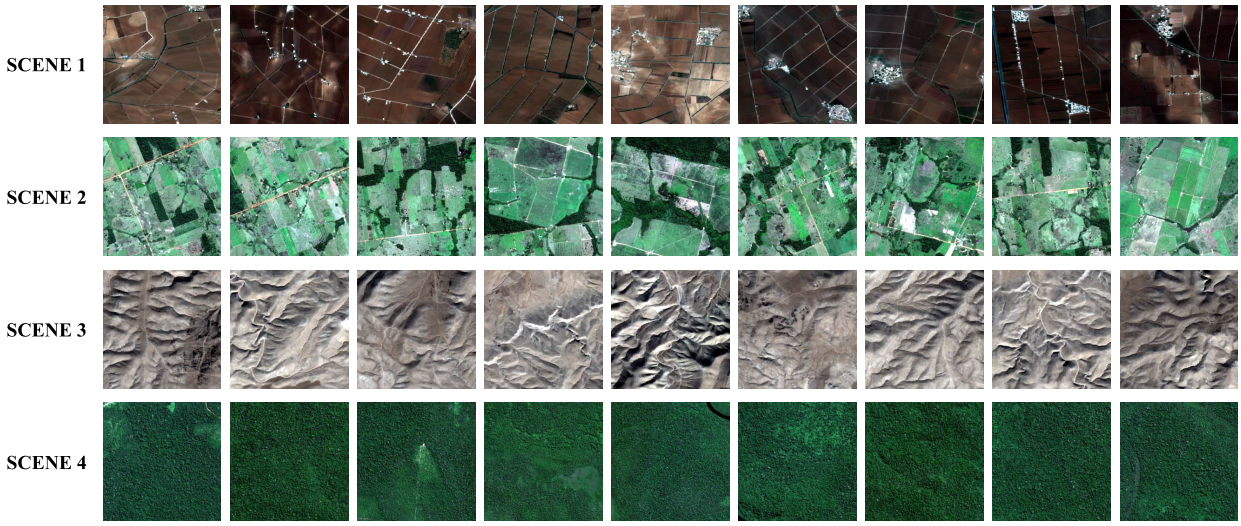


Fig. 1. In remote sensing datasets, images are typically cropped from large scene images. Images cropped from the same scene exhibit certain similarities in terms of color, texture details, and overall layout.

randomly distributed throughout the entire image. However, CL, as a global discriminant task, excels in extracting global discriminative information and inherently struggles to capture details. This inherent challenge makes it difficult for CL to capture the fine-grained details essential for remote sensing tasks.

To overcome these limitations, we propose SwiMDiff, a novel self-supervised pretraining framework tailored for RSIs. SwiMDiff enhances CL by incorporating a scene-wide matching strategy. This strategy recalibrates data based on the associated scenes to avoid mislabeling samples as negatives. It incorporates consideration for intraclass similarity, enhancing the model's ability to extract global discriminative and semantic information. Furthermore, by integrating a diffusion model, SwiMDiff strengthens the CL encoder, creating a unified training framework that synergizes both techniques. Empowered by pixel-level diffusion constraints, SwiMDiff places a greater emphasis on local and detailed information. SwiMDiff not only captures global semantic information based on CL but also pays more attention to fine-grained features, enabling it to capture and provide richer information for downstream tasks in the remote sensing domain.

Our evaluation of SwiMDiff involves change detection and land-cover classification tasks on datasets like OSCD [30] and LEVIR-CD [31] for change detection, and BigEarthNet [32] and EuroSAT [33] for land-cover classification. The results demonstrate SwiMDiff's superior performance, indicating its potential promising applicability in the realm of remote sensing.

In summary, this article makes the following contributions.

1) The development of SwiMDiff, an innovative self-supervised pretraining framework for RSI analysis. This framework is pioneering in its integration of the diffusion model with CL training, establishing a new benchmark in effective pretraining methodologies for RSI.
2) The implementation of a scene-wide matching strategy within the SwiMDiff framework. This approach capitalizes on the semantic similarities of images from identical geographical scenes. It significantly enhances SwiMDiff's ability to accurately interpret and analyze RSIs.
3) Demonstrated excellence in practical applications. SwiMDiff has been rigorously tested on change detection tasks using the OSCD and LEVIR-CD datasets and on land-cover classification tasks with the BigEarthNet and EuroSAT datasets. In these applications, SwiMDiff has achieved state-of-the-art results, showcasing its exceptional capacity to handle a variety of complex remote sensing tasks.

II. Related Work

A. Diffusion Model

Diffusion models, a subset of probabilistic denoising networks, operate by introducing noise into data, then learning to reverse this process to generate refined samples [34]. A notable subclass of these models, the denoising diffusion probabilistic models (DDPMs) [35], encompass forward and reverse mechanisms. In the forward process, random noise is incrementally introduced to an image, functioning like a Markov chain where each step depends only on its predecessor. Over time, this added noise transforms the original image into a purely noisy state. Conversely, the reverse process uses the noise patterns from the forward process to reconstruct the original image. This is achieved by feeding the noisy image into a neural network, which is trained to predict the noise, constrained by the ℓ_2-norm between the predicted and actual noise.

Diffusion models have rapidly advanced in recent years, showing exceptional promise in fields like computer vision [36], [37], [38]. Their generative capabilities have been leveraged for robust feature extraction and representation. For instance, Xiang et al. [39] utilized the networks within diffusion models for self-supervised feature extraction. In remote sensing, Bandara et al. [40] applied DDPM for SSL on RSIs, leading to applications in change detection.

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Another application in remote sensing is the BSDM approach, which employs diffusion models' denoising and representational strengths for background suppression in hyperspectral images, facilitating hyperspectral anomaly detection tasks [41]. Hence, we consider the diffusion model for self-supervised representation learning. However, relying solely on it demands substantial data, a high-parameter network, and extensive iterations [39], struggling to capture global semantic information. Therefore, we propose leveraging the diffusion model to assist CL, enhancing the encoder for more accurate advanced semantics and richer fine-grained details.

B. False Negative Samples (FNSs) in CL

Since Hadsell et al. [42] introduced the CL paradigm, it has continuously been optimized and improved, centering around the core concept of positive and negative samples. CMC leverages varying views of the same object to enhance mutual information [43]. MoCo [24] and SimCLR [27] distinguish augmented samples within momentum-updated queues and memory banks, respectively, for the pretext task of instance discrimination, significantly enhancing the efficacy and practicality of CL. Conventionally, different augmentations of the same image are considered positive pairs, while different images are treated as negatives. However, this straightforward classification can lead to the creation of FNSs, where similar-category samples are erroneously marked as negatives. This mislabeling causes sample confounding, impairing model accuracy [29].

Some studies eliminate negative samples. For example, BYOL [26] and SimSiam [44] solely utilize the similarity between positive pairs for predictive tasks, and Barlow Twins aims to approximate the cross correlation matrix generated from positive pairs to the identity matrix [45]. However, their effectiveness has not seen substantial improvement, failing to fundamentally resolve the issue of FNS. Other studies focus on redefining sample relationships, and effectively addressing the FNS problem. ASCL, another innovative method, adapts the traditional instance discrimination task into a multiinstance format by dynamically relabeling FNSs, thereby improving the overall efficacy of CL [46]. FALSE method tackles FNSs in RSIs by adjusting their influence in the loss function, based on coarse and precise assessments [29]. However, traversing the entire database for FNSs is resource-intensive and lacks close correlation with positive samples, resulting in suboptimal feature extraction. In this article, our proposed SwiMDiff precisely locates and recalibrates FNSs within the same scene subset, significantly enhancing precision and efficiency in acquiring FNSs during CL.

C. SSL in Remote Sensing

The advancement and proliferation of airborne and satellite optical sensors have made remote sensing imagery more accessible than ever. This ease of access is further supported by large-scale remote sensing datasets, such as SEN12ms [47] and fMoW [48], and platforms like Google Earth Engine [49], which facilitate the creation of customized datasets. These developments are significantly advancing SSL in the remote sensing field.

Remote sensing data's distinct characteristics have led to the development of specialized SSL methods. For instance, Ayush et al. [50] have utilized the spatiotemporal nature of RSIs to introduce a geographic location classification task in CL, effectively merging self-supervised and supervised learning techniques. SauMoCo has been developed to enhance semantic diversity by exploiting the semantic similarities between adjacent tiles in RSIs [51]. SeCo takes advantage of the seasonal consistency in RSIs to maximize the use of satellite imagery [52]. In addition, STICL employs optimal transport techniques to handle RSIs across different spatial–temporal scenes, aiming to learn spatial–temporal invariant representations [53].

III. Method

We propose SwiMDiff, a novel self-supervised pretraining framework for RSIs, combining the strengths of CL and generative learning. SwiMDiff aims to refine an encoder for enhanced feature extraction in various downstream tasks. As depicted in Fig. 2, SwiMDiff comprises two main components: a dual-branch CL network (Section III-A) and a diffusion model network (Section III-B).

A. Scene-Wide Matching CL

To enhance representation extraction for RSIs, we introduce a scene-wide matching approach within the CL framework. This method builds upon the foundation of CL and incorporates the concept of scene-wide matching. The network architecture is similar to the original CL network, consisting of dual-branch momentum-updated encoders and projection heads. Our novel contribution lies in the embedding space, specifically recalibrating the negative sample set in the dictionary for more accurate loss computation.

1) Contrastive Learning: An anchor image is subjected to different augmentations T_0 and T_1, such as random cropping, color jittering, and flipping [23], generating query view x^q and key view x^k for the CL network. The two views are processed by the network encoder f_q and the momentum encoder f_k, yielding features $q = f_q(x^q)$ and $k = f_k(x^k)$. Then, the projection heads h_q and h_k transform high-dimensional features into the representations $z_q = h_q(q)$ and $z_k = h_k(k)$ in a spherical space. The original loss seeks to minimize the distance between z_q and z_k while concurrently maximizing the separation from the dictionary queue. And we unify all samples as $\{z_k, z_1, \ldots, z_n\} \triangleq \{z_0, z_1, \ldots, z_n\}$

$$L_{\text{ori}} = -\log \frac{\exp\left(z_q^\top z_k/\tau\right)}{\exp\left(z_q^\top z_k/\tau\right) + \sum_{i=1}^{n} \exp\left(z_q^\top z_i/\tau\right)} = -\sum_{i=0}^{n} l_i \log \frac{\exp\left(z_q^\top z_i/\tau\right)}{\sum_{j=0}^{n} \exp\left(z_q^\top z_j/\tau\right)} \tag{1}$$

where

$$l_i = \begin{cases} 1, & i = 0 \\ 0, & \text{otherwise.} \end{cases}$$

Here $z_q^\top$ represents the transpose of z_q, and $\{l_0, l_1, \ldots, l_n\}$ refers to the one-hot pseudo label showing that only x^k belong to the same class as x^k.

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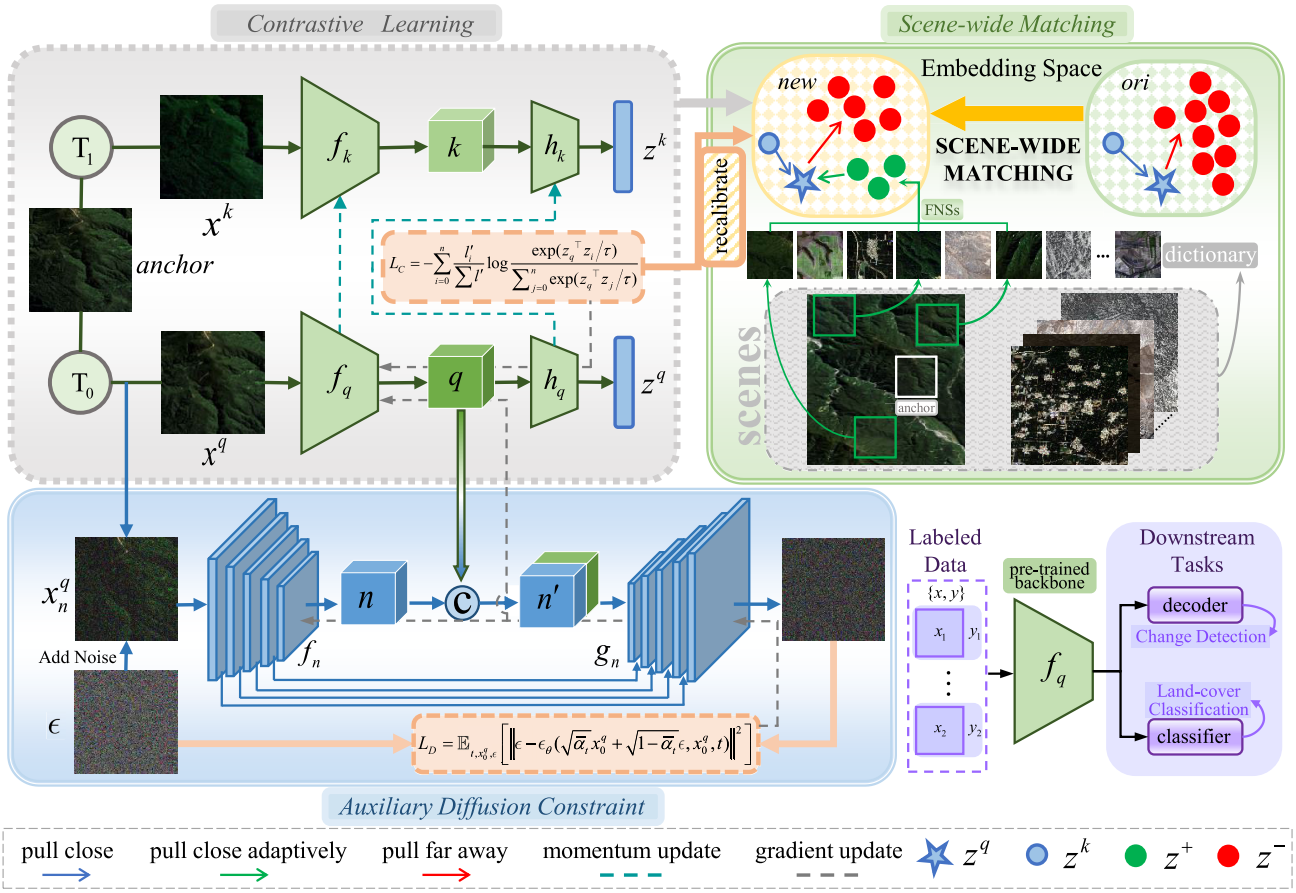


Fig. 2. Diagram of the SwiMDiff. The network architecture is bifurcated into two components: 1) a dual-branch structure for CL and 2) a diffusion model network comprising an encoder and a decoder.

2) Scene-Wide Matching: As illustrated in Fig. 2, images under a broad scene exhibit varying degrees of semantic similarity and intraclass relationships. We identify images belonging to the same scene with the anchor as FNSs. Initially, we select FNSs from the dictionary bank and constitute a false negative set $Z_{\text{FNS}} = \{z_1^+, z_2^+, \ldots, z_m^+\}$. Conversely, the remaining samples are still maintained as negatives z^-. Given the intricate and multifaceted nature of the surface environment, each sample from Z_{FNS} encompasses pertinent but not identical geographical information. Thus, we utilize an adaptive soft mechanism [46] to reformulate FNSs.

Based on the cosine similarity $\{d_1, d_2, \ldots, d_m\}$ between FNSs and z_0, the relative distribution $\{b_1, b_2, \ldots, b_m\}$ between z_0 and other representations in Z_{FNS} can be considered as soft labels as

$$b_i = \frac{\exp(d_i/\tau')}{\sum_{j=1}^{m} \exp(d_i/\tau')}, \quad i = 1, \ldots, m. \tag{2}$$

Subsequently, the adaptive soft labels are derived as $\{s_1, s_2, \ldots, s_m\}$ and limited less than one for keeping the positive itself z_0 still the most confident sample

$$s_i = \min\left(1, b_i * \left[1 - \frac{H(\mathbf{b})}{\log(n)}\right]\right), \quad i = 1, \ldots, m \tag{3}$$

where $H(\mathbf{b})$ represents the Shannon entropy. The scene-wide matching labels are relabeled as

$$l_i' = \begin{cases} 1, & i = 0 \\ s_i, & i \neq 0 \cap z_i \in Z_{\text{FNS}} \\ 0, & \text{otherwise.} \end{cases}$$

Here $\{l_0', l_1', \ldots, l_n'\}$ indicates z_0 as the most confident positive. Other FNSs from the same scene are adaptively assigned their respective label values based on varying similarities. Conversely, other samples are still considered as unrelated negatives. Ultimately, we employ the normalized l' to obtain the scene-wide matching contrastive loss

$$L_C = -\sum_{i=0}^{n} \frac{l_i'}{\sum l'} \log \frac{\exp\left(z_q^\top z_i/\tau\right)}{\sum_{j=0}^{n} \exp\left(z_q^\top z_j/\tau\right)}. \tag{4}$$

Through the contrastive loss L_C, the network encoder f_q is trained to distill global discriminative and deep semantic information.

B. Auxiliary Diffusion Constraint

To extract finer details from RSIs for downstream tasks, we integrate a diffusion model as an auxiliary task, adding a diffusion constraint to f_q for iterative optimization. We adopt

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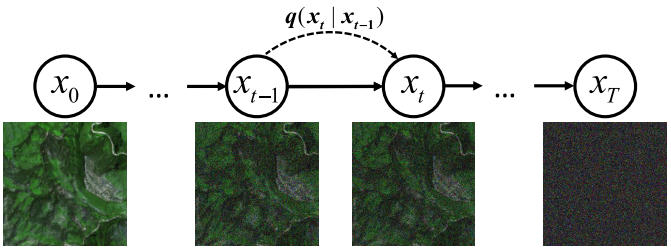


Fig. 3. Forward diffusion process of the diffusion model. It's similar to a Markov chain, where noise is added based on the previous state.

the diffusion model's training mechanism, predicting noise introduced during the forward process with both encoder f_n and decoder g_n.

1) Gaussian Forward Diffusion: We first add random Gaussian noise $\epsilon \sim \mathcal{N}(0, \boldsymbol{I})$ to the clean image x^q (i.e., x_0^q) t times cumulatively. As depicted in Fig. 3, the noising operation at step t solely related to step $t-1$ is defined as follows:

$$q\left(x_t^q \middle| x_{t-1}^q\right) = \mathcal{N}\left(x_t^q; \sqrt{1-\beta_t} x_{t-1}^q, \beta_t \boldsymbol{I}\right) \tag{5}$$

where $(x_0^q, x_1^q, \ldots, x_T^q)$ represents a Markov chain of noisy images and $(\beta_1, \beta_2, \ldots, \beta_t)$ denotes the coefficients linearly interpolated between 0.0001 and 0.02, governing the noise variance at each step. By accumulating noise over t iterations and reparameterizing the intermediate process, we then obtain the distribution of the noising image as

$$\begin{aligned} q\left(x_t^q \middle| x_0^q\right) &= \prod_{i=1}^{t} q\left(x_i^q \middle| x_{i-1}^q\right) \\ &= \mathcal{N}\left(x_t^q; \sqrt{\bar{\alpha}_t} x_0^q, \left(1-\bar{\alpha}_t\right) \boldsymbol{I}\right) \end{aligned} \tag{6}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. With t growing, x_t gets closer to pure Gaussian distribution. Based on (6), we derive x_n^q by adding random n step noise to x_0^q.

2) Training Mechanism: The obtained noised image x_n^q is used as input for the diffusion model's U-Net network. Through the encoder f_n, it is mapped to high-dimensional features $n = f_n(x_n^q)$. Notably, the two networks are integrated by concatenating feature q emerged from the encoder f_q of CL and feature n into n', which is subsequently fed into the decoder g_n. The dimensions of the two feature maps are consistent, achieving integration at the feature level.

The feature q retains the global discriminative information and high-level semantic content of the clean image. It serves as guiding information for the diffusion model, facilitating noise prediction. Guided by q, both f_n and g_n strive to predict the noise added to x_0^q. We condition the step estimation function ϵ with a noise-free image prior, denoted as

$$\begin{aligned} \epsilon_\theta\left(x_n^q, x_0^q, t\right) &= g_n\left(\left(f_n\left(x_n^q, t\right) \oplus f_q\left(x_0^q\right)\right), t\right) \\ &= g_n((n \oplus q), t) \\ &= g_n\left(n', t\right) \end{aligned} \tag{7}$$

where $\oplus$ denotes the concatenation operation of two feature matrices. Furthermore, f_q, f_n and g_n are trained to minimize the pixel-level distance between the original noise ϵ_θ and the predicted noise

$$L_D = \mathbb{E}_{t, x_0^q, \epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t} x_0^q + \sqrt{1-\bar{\alpha}_t}\epsilon, x_0^q, t\right)\right\|^2\right]. \tag{8}$$

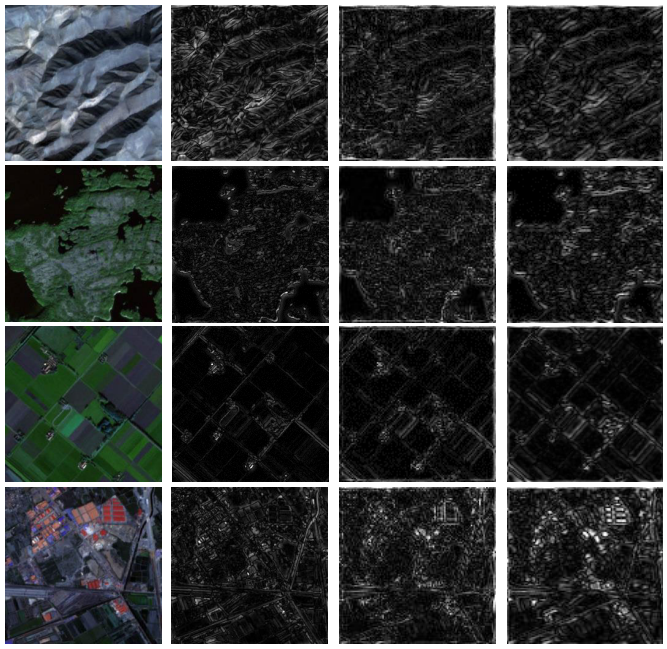


Fig. 4. High-frequency components of the images and its shallow features. (a) Input images. (b) High-frequency components extracted from input images. (c) High-frequency details of shallow features extracted from encoder pretrained by CL. (d) High-frequency details of shallow features extracted from encoder pretrained by model integrating with the diffusion model.

Currently, integrating CL and the diffusion model essentially establishes a multitask learning framework. The addition of diffusion model-based pixel-level constraints renders the participating encoder f_q to receive richer and more detailed supervisory signals. This results in the preservation and emphasis of shallow features, such as edges and textures, as shown in Fig. 4. We represent the fine-grained information contained in features by extracting high-frequency components. Compared to the original image, the features after integrating the diffusion model exhibit clearer boundaries and richer textural details. Fig. 4 intuitively demonstrates that the diffusion model, serving as an auxiliary branch, can significantly reduce information loss in CL, providing a more effective and comprehensive feature representation for downstream tasks.

C. Joint Training

The overall objective of SwiMDiff is formulated as

$$L = \lambda_C * L_C + \lambda_D * L_D. \tag{9}$$

Here, λ_C and λ_D are weight factors balancing the impacts of the two losses. SwiMDiff not only enhances discriminability among samples but also preserves detailed features within images. The scene-wide matching CL incorporates similarity information between adjacent tiles, improving high-level discriminative feature extraction. The diffusion constraint enables

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the encoder to grasp the data's inherent structural and distributional nuances, focusing on pixel-level details.

IV. Experimental Results

We assess the representations produced by our method across two downstream tasks: change detection task and land-cover classification task.

A. Self-Supervised Pretraining

1) Pretraining Dataset: We utilize a subset of the Sen12MS [47] dataset for pretraining, considering time constraints. The Sen12MS dataset comprises 1 80 662 triplets of Sentinel-1 SAR data, Sentinel-2 [54] multispectral images, and MODIS maps. The Sentinel-1 SAR data contains two spectral bands, while the Sentinel-2 multispectral image is comprised of ten spectral bands. For our pretraining, we randomly select 10 000 atmospherically corrected Sentinel-2 RGB images (256 × 256) from each season, with 85% allocated for training and 15% allocated for validation.

2) Pretraining Implementation Details: Our framework integrates the scene-wide matching method with **Momentum Contrast** (MoCo-v2) [24] baseline and employs the **DDPM** [35] framework for the diffusion model. We use ResNet-18 [55] as our encoders and a two-layer MLP as the projection head. The model is pretrained for 1000 epochs with a batch size of 256 distributed across 4 Nvidia A100 GPUs. For the CL network, we employ an SGD optimizer, setting the initial learning rate at 0.03, the momentum at 0.9, and the weight decay at 1e-4. Meanwhile, the diffusion model network is trained using an Adam optimizer [56] with a learning rate of 1e-3. The joint training involves weight factors λ_C of 1 and λ_D of 10. In addition, we set the temperature scaling parameters τ and τ' at 0.1 and 0.05, respectively, in the contrastive loss component.

3) Comparative Methods: We compare our proposed SwiMDiff with several methods, consisting of random initialization and some self-supervised pretraining. The self-supervised pretraining methods include MoCo-v2 [24] (baseline), Barlow Twins [45] (eliminating negative samples and reducing redundancy), DiRA [57] (uniting discriminative, restorative, and adversarial learning in a unified manner), and tri-SimCLR [58] (introducing a three-factor contrastive loss).

In addition, our evaluation extends to ablation studies focusing on specific components of our framework. This includes an examination of the impact of scene-wide matching when added to the baseline MoCo-v2 (termed MoCo-v2 + SwiM) and an analysis of the joint pretraining approach that combines CL with the diffusion model (denoted as MoCo-v2 + Diff). Each of these methods, along with our proposed SwiMDiff, is trained on the same 10 000-sample subset of the Sen12MS dataset.

4) Metrics in Downstream Tasks: We opt for change detection and land-cover classification as our downstream tasks, which are both significant and widely examined in remote sensing [4], [5], [52]. These two tasks provide different and valuable perspectives for evaluating the pretrained model roundly, one emphasizing shallow-level details and the other focusing on deep-level semantics.

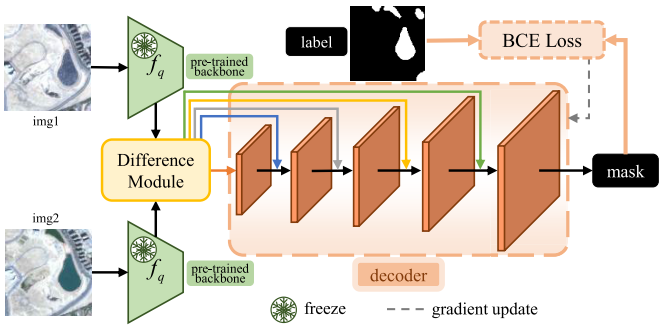


Fig. 5. Network architecture for the change detection task. The images taken at different times are first processed by a pretrained and frozen encoder f_q. It extracts two sets of features from these images. These feature sets are then passed through a different module. The resulting difference is then input into the decoder for further processing.

We employ various mathematical metrics to assess performance in change detection and land-cover classification. The outcomes of these downstream tasks reflect the performance of models obtained through different self-supervised methods.

a) Metrics in change detection: In change detection, we utilize the $F1$ score to assess the consistency and discrepancies between the output mask and the ground truth. $F1$ score is a comprehensive evaluation metric that quantitatively analyzes each pixel, represented as the harmonic average value of precision and recall [5]

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{10}$$

where

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{12}$$

Here, TP represents the number of truly positive samples correctly classified as positive, FP represents the number of truly negative samples incorrectly classified as positive, and FN represents the number of actually positive samples incorrectly classified as negative.

b) Metrics in land-cover classification: For land-cover classification tasks, we adopt the mean average precision (mAP) to measure the quality of classification results. AP represents the average precision for a single class label, and its value corresponds to the area under the precision–recall curve. And mAP stands for the average of the individual AP values calculated for all N categories

$$\text{mAP} = \frac{\text{AP}}{N} = \frac{\int_0^1 p(r)dr}{N} \tag{13}$$

where p refers to precision and r denotes recall [8].

B. Change Detection on Onera Satellite

This task focuses on detecting changes in image pairs captured from the same geographic location but at different times. We evaluate the performance using the $F1$ score and utilize the Onera satellite change detection (OSCD) [30] dataset. This dataset comprises 24 pairs of multispectral images across

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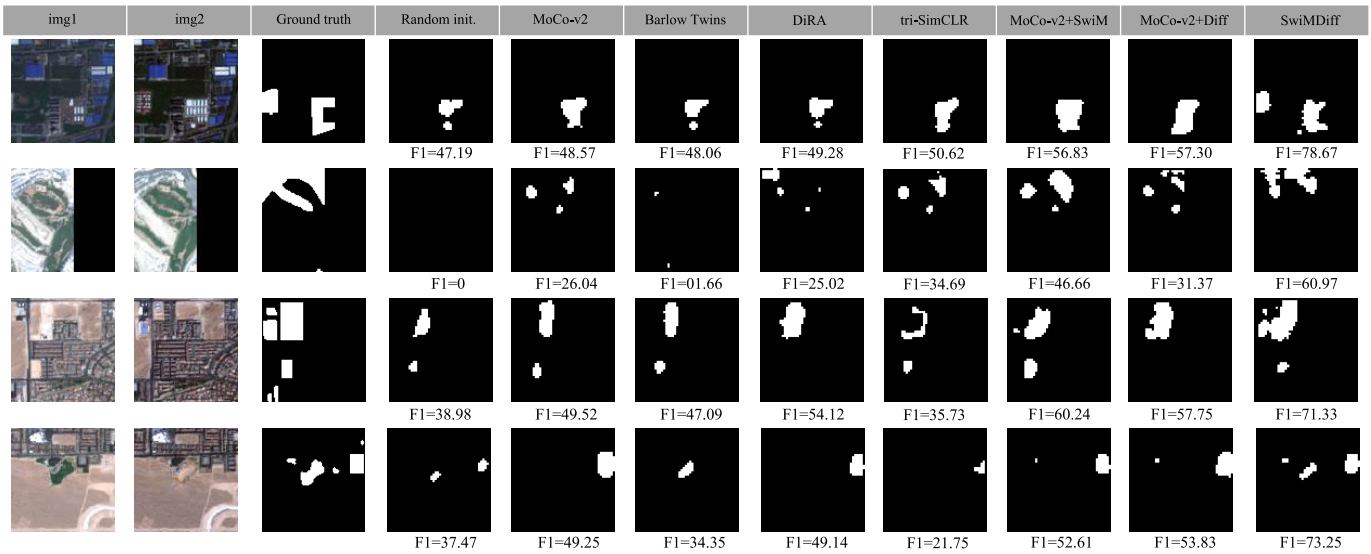


Fig. 6. Comparison and ablation of visualization results on the OSCD task.

TABLE I

COMPARISON RESULTS OF PRECISION, RECALL, AND $F1$ (%) ON THE CHANGE DETECTION OF OSCD VALIDATION SET

Method	Precision(%)	Recall(%)	F1(%) ↑
Random init.	69.8	23.5	34.1
MoCo-v2	54.4	40.7	45.6
Barlow Twins	59.8	35.7	44.2
DiRA	63.5	37.9	46.6
tri-SimCLR	62.3	37.6	46.2
SwiMDiff(ours)	63.6	40.9	**49.6**

13 spectral bands of Sentinel-2, with resolutions of 10, 20, and 60 m. For each pair of multitemporal images, there exists a corresponding change detection label, with white pixels indicating changes and black pixels representing no change. Our experiments are confined to the RGB bands, and the dataset is divided for training and validation following the methodology established by Daudt et al. [30].

1) Implementation Details: The network setup for this task, as shown in Fig. 5, involves processing each pair of images into two sets of features using the pretrained ResNet-18 [55] backbone. The absolute difference between these feature sets is then calculated and input into a U-Net [59] decoder to create change detection masks. For this task, we keep the ResNet-18 backbone static and focus on training the U-Net's remaining components for 100 epochs, using a batch size of 32. In line with the suggestions of Manas et al. [52], our approach includes image augmentation through random horizontal flips and 90° rotations. We use an Adam optimizer [56] with a weight decay of 1e-4 and an initial learning rate of 1e-3.

2) Results Discussion:

a) Comparison results: As presented in Table I and Fig. 6, we evaluate SwiMDiff against various methods, including random initialization, MoCo-v2 [24] (our baseline), Barlow Twins [45], DiRA [57], and tri-SimCLR [58].

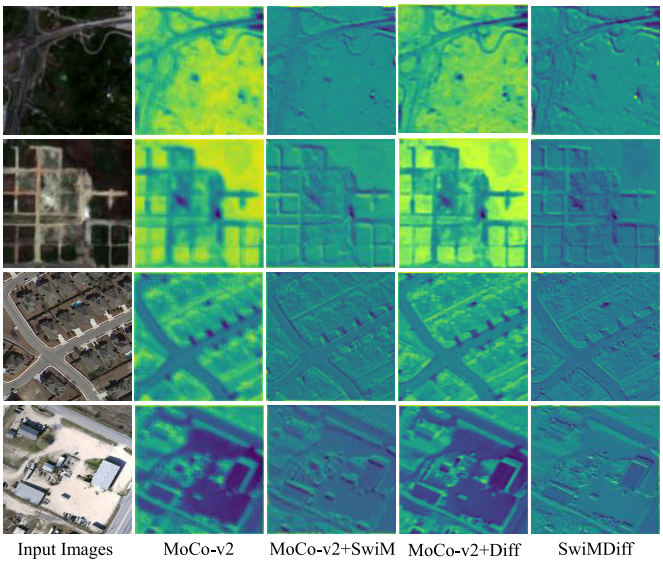


Fig. 7. Qualitative results of our experiments. The top two rows are from OSCD and the bottom two rows are from LEVIR-CD. It can be seen that the extracted features after pretraining of different methods are all enhanced on the baseline.

SwiMDiff achieved the highest $F1$ score, improving by 4.0% over the baseline, quantitatively demonstrating its outstanding performance in change detection tasks. We also visualize some qualitative examples, further highlighting the effectiveness of SwiMDiff in change detection on the OSCD dataset. Specifically, as shown in the top two rows of Fig. 7, SwiMDiff can extract more precise image details and sharper object boundaries compared to the baseline's shallow features. These improvements emphasize our method's ability to extract shallow pixel-level features and image detail information. In addition, from the visual results in Fig. 6, for tiny buildings in low-resolution OSCD dataset, the detection accuracy of our SwiMDiff enjoys superiority over other comparative methods. Its results are closer to the ground truth in terms of

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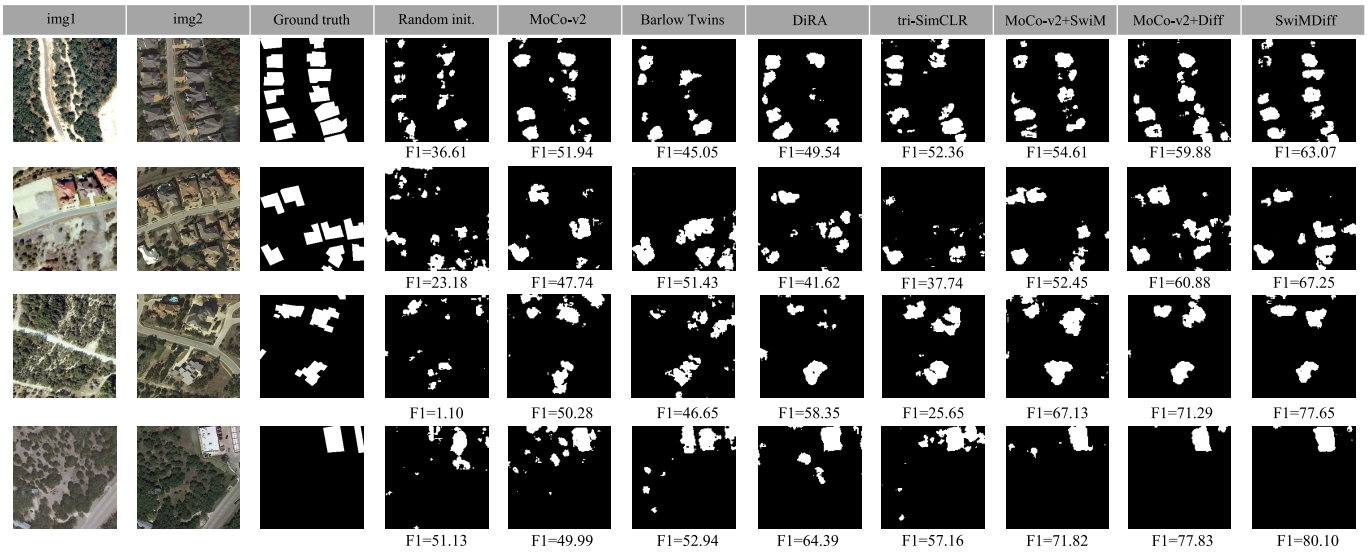


Fig. 8. Comparison and ablation of visualization results on the LEVIR-CD change detection task.

TABLE II
ABLATION RESULTS OF PRECISION, RECALL, AND $F1$ (%) ON THE CHANGE DETECTION OF OSCD AND LEVIR-CD VALIDATION SETS

Components		OSCD			LEVIR-CD		
SwiM	Diff	Precision(%)	Recall(%)	F1(%)↑	Precision(%)	Recall(%)	F1(%)↑
		54.4	40.7	45.6	81.7	76.8	79.1
✓		50.2	45.1	46.7	82.7	77.5	80.0
	✓	61.8	41.3	48.6	82.9	77.6	80.1
✓	✓	63.6	40.9	**49.6**	83.6	78.3	**80.9**

contours and shapes, further demonstrating SwiMDiff's ability to extract key semantic information from images.

b) Ablation results: The ablation study results, presented in Table II and Fig. 6, demonstrate the enhancements achieved by adding single modules to the baseline. The inclusion of scene-wide matching (MoCo-v2 + SwiM) improves the $F1$ score by 1.1%, while integrating the diffusion model (MoCo-v2 + Diff) leads to a 3.0% increase. As evident in Fig. 7, features refined through the diffusion auxiliary task are markedly clearer and more defined compared to the baseline. Furthermore, the MoCo-v2 + SwiM module also shows an improvement in the extraction of detailed features. This aligns with our hypothesis that incorporating a diffusion constraint deepens the understanding of detail and pixel-level information in images. SwiMDiff, by combining these two modules, significantly enhances the original self-supervised method, resulting in more nuanced and fine-grained change detection.

C. Change Detection on LEVIR-CD

LEVIR-CD [31] is a comprehensive remote sensing change detection dataset that includes bitemporal image pairs from 20 diverse regions in Texas, USA, spanning 5–14 years. This dataset encapsulates a wide array of buildings, such as villas residences, tall apartments, small garages, and large warehouses. It comprises 637 very high-resolution image pairs from Google Earth [49], each with dimensions of 1024 × 1024 pixels, meticulously inspected to ensure quality.

1) Implementation Details: For this task, we implement the same architectural model as used in the change detection on the Onera satellite task (Section IV-B1). Our approach utilizes a pretrained, frozen ResNet-18 [55] to extract features. The training is conducted over 100 epochs with a batch size of 32, using an Adam optimizer [56] with an initial learning rate of 1e-3 and a weight decay of 1e-4. We follow the data partitioning strategy suggested by Chen and Shi [31] for training and validation, segmenting the images into 256 × 256 nonoverlapping tiles and disregarding completely black labels.

2) Results Discussion:

a) Comparison results: As indicated in Table III, SwiMDiff outperforms other methods in precision, recall, and $F1$ score, demonstrating a notable 1.8% improvement in $F1$ score over the baseline. This underlines its superior performance in change detection. In Fig. 7, the bottom two rows clearly show that SwiMDiff extracts more detailed and sharper features compared to the baseline, which are crucial for accurate image analysis. Fig. 8 shows the detection masks generated by different methods on the validation set. As shown in Fig. 8, for tiny buildings, our SwiMDiff method can

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TABLE III
COMPARISON AND ABLATION RESULTS OF PRECISION, RECALL, AND $F1$ (%) ON THE CHANGE DETECTION OF LEVIR-CD VALIDATION SET

Method	Precision(%)	Recall(%)	F1(%) ↑
Random init.	67.3	44.1	53.1
MoCo-v2	81.7	76.8	79.1
Barlow Twins	80.1	75.3	77.5
DiRA	83.5	75.0	79.0
tri-SimCLR	83.2	75.9	79.5
SwiMDiff(ours)	83.6	78.3	**80.9**

detect more changed pixels compared to other approaches. In addition, SwiMDiff's advantages in semantic enhancement and highlighting details can mitigate miss-recognition caused by noise interference and pseudo changes from lighting differences and occlusions, thus producing the best visual representation. The fewer omission faults and commission errors highlight SwiMDiff's advancement in change detection performance, both qualitatively and quantitatively, across diverse datasets and scenes with sophisticated structures.

b) Ablation results: The ablation study, detailed in Table II and Fig. 8, reveals that MoCo-v2 + SwiM and MoCo-v2 + Diff, respectively, improve $F1$ scores by 0.9% and 1.0% compared to the baseline. This demonstrates the effectiveness of each module when applied independently. Notably, the scene-wide matching module also shows improvements in extracting detailed features and accomplishing pixel-level tasks. SwiMDiff, combining these two modules, further refines feature extraction, enhancing the overall performance in change detection.

D. Land-Cover Classification on BigEarthNet

BigEarthNet [32], a comprehensive dataset, consists of 590 326 distinct multispectral images from Sentinel-2 [54], sourced from 125 tiles covering ten European countries. Designed for multilabel land-cover classification, it encompasses 19 diverse categories, with each image annotated with multiple classes. Covering 1.2 × 1.2 km/image at resolutions of 10, 20, and 60 m per pixel, we selectively use 10% of the dataset, excluding images fully obscured by snow, clouds, or shadows, for training and validation.

1) Implementation Details: Our classification task utilizes a pretrained ResNet-18 [55] as the backbone, coupled with an additional MLP layer for classifying high-dimensional features. The network is trained for 100 epochs with a batch size of 256. Specifically, as demonstrated in Fig. 9, for the linear probing approach, we freeze the backbone and employ an Adam optimizer [56] with an initial learning rate set to 1e-3, focusing on iteratively optimizing the classifier. For the fine-tuning approach, we adjust the entire network including the backbone and classifier with an initial learning rate of 1e-5. The learning rate is reduced by 10 at 60% and 80% of the total epochs.

2) Results Discussion:

a) Comparison results: As depicted in Table IV, SwiMDiff significantly outperforms other methods in pretraining accuracy, particularly evident with a 1.3% increase in linear probing performance over the baseline. Notably, when the entire network is fine-tuned, performance differences among methods are less pronounced. In the fine-tuning phase, SwiMDiff exhibits an increment of just 0.7% observed. This underscores SwiMDiff's enhanced ability to extract global discriminative information and high-level semantic features.

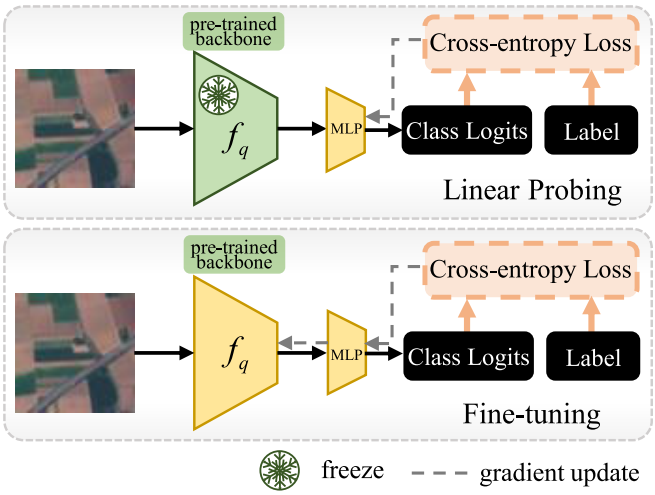


Fig. 9. Network architecture for a land-cover classification task. We load pretrained weights as the initial weights for the encoder f_q. In the case of linear probing, the encoder's weights are frozen, while only the MLP layer used for classification is updated with gradients. In the case of fine-tuning, both the encoder and the classifier are updated with gradients.

TABLE IV
COMPARISON RESULTS OF MAP ON THE LAND-COVER CLASSIFICATION OF BIGEARTHNET VALIDATION SET

Method	Linear Probing(%) ↑	Fine-tuning(%) ↑
Random init.	43.6	69.4
MoCo-v2	68.6	80.4
Barlow Twins	61.1	79.2
DiRA	69.7	80.7
tri-SimCLR	67.5	79.6
SwiMDiff(ours)	**69.9**	**81.1**

b) Ablation results: Table V details the impact of individual modules. Compared to the baseline, the accuracy of linear probing with MoCo-v2 + SwiM increases by 1.1%, while the fine-tuning result makes a mere 0.3% improvement. For MoCo-v2 + Diff, there is a 1.2% enhancement in linear probing accuracy and a 0.5% increase for fine-tuned outcomes. Both modules individually contribute to the elevation of detection precision. The scene-wide matching technique and the auxiliary task of the diffusion model both serve to bolster the model's ability to learn global semantic features.

c) Ablation on weight factors: To achieve optimal training results for the SwiMDiff, we design ablation experiments for the values of λ_C and λ_D. We maintain the weight of CL loss at 1 while varying the weight of the diffusion loss during the training process. We test and evaluate pretrained models with different weight ratios on the classification task of BigEarthNet with linear probing, resulting in Table VI. We find

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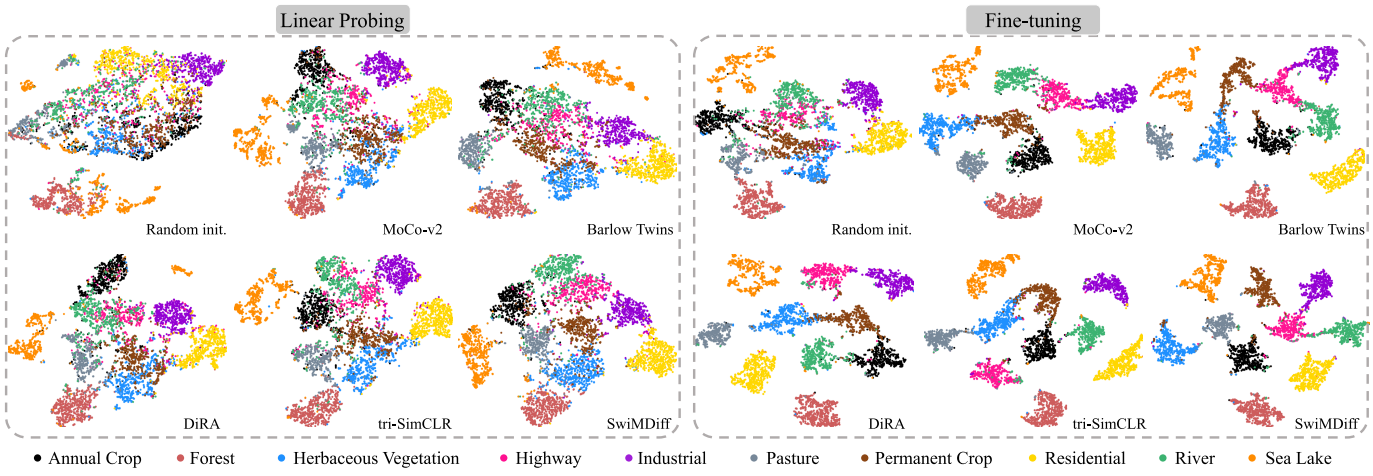


Fig. 10. t-SNE visualization of learned representations on the validation set of EuroSAT.

TABLE V
ABLATION RESULTS OF MAP ON THE LAND-COVER CLASSIFICATION OF BIGEARTHNET AND EUROSAT VALIDATION SETS

Components		BigEarthNet		EuroSAT	
SwiM	Diff	Linear Probing(%)↑	Fine-tuning(%)↑	Linear Probing(%)↑	Fine-tuning(%)↑
		68.6	80.4	86.6	94.8
✓		69.7	80.7	88.5	95.7
	✓	69.8	80.9	88.3	95.4
✓	✓	**69.9**	**81.1**	**89.1**	**96.1**

TABLE VI
ABLATION EXPERIMENTS ON DIFFERENT WEIGHT COEFFICIENTS DURING THE PRETRAINING PHASE

λ_C/λ_D	$1/14$	$1/12$	$1/10$	$1/8$	$1/6$
Accuracy(%)	69.3	69.8	**69.9**	69.8	69.5

that the weight coefficient ratio is very crucial during the pretraining phase. As shown in Table VI, the best integration effect of CL and diffusion model occurs when the weight coefficients are 1 and 10, respectively. However, when the weight coefficient ratio is too large or too small, it is not conducive to the integration of them.

E. Land-Cover Classification on EuroSAT

EuroSAT [33] is a single-label land-cover classification dataset, featuring images with dimensions of 64 × 64 pixels. Originating from the Sentinel-2 satellite, it encompasses imagery across 13 spectral bands. Within EuroSAT, there are ten distinct classes, each containing between 2000 and 3000 images, summing up to a total of 27 000 images. Noteworthy, the dataset is curated to exclude images with high cloud coverage but does not undergo atmospheric correction. We partition the data into training and validation sets following the method proposed in [60].

1) Implementation Details: We attach a linear classification head after the pretrained backbone and train it for 100 epochs with a batch size of 256. The training process, as shown in Fig. 9, is divided into linear probing and fine-tuning phases, each with respective initial learning rates of 1e-3 and 1e-5, reduced by 0.1 at the 60% and 80% epochs.

TABLE VII
COMPARISON RESULTS OF MAP ON THE LAND-COVER CLASSIFICATION OF EUROSAT VALIDATION SET

Method	Linear Probing(%) ↑	Fine-tuning(%) ↑
Random init.	69.5	85.8
MoCo-v2	86.6	94.8
Barlow Twins	86.8	95.0
DiRA	87.5	95.2
tri-SimCLR	88.6	95.9
SwiMDiff(ours)	**89.1**	**96.1**

2) Results Discussion:

a) Comparison results: As shown in Table VII, SwiMDiff achieved the best classification results on the Eurosat dataset, improving by 2.5% over MoCo-v2 in linear probing and by 1.3% in fine-tuning. Figs. 10 and 11, respectively, display the t-SNE clustering visualization and the confusion matrix for the classification results on the EuroSAT validation set. In Fig. 10, compared to MoCo-v2, SwiMDiff demonstrates a more distinct and effective ability to cluster the same categories and separate different ones. For instance, in linear probing, "Herbaceous Vegetation," "Pasture," and "River" are each more tightly clustered, while in fine-tuning,

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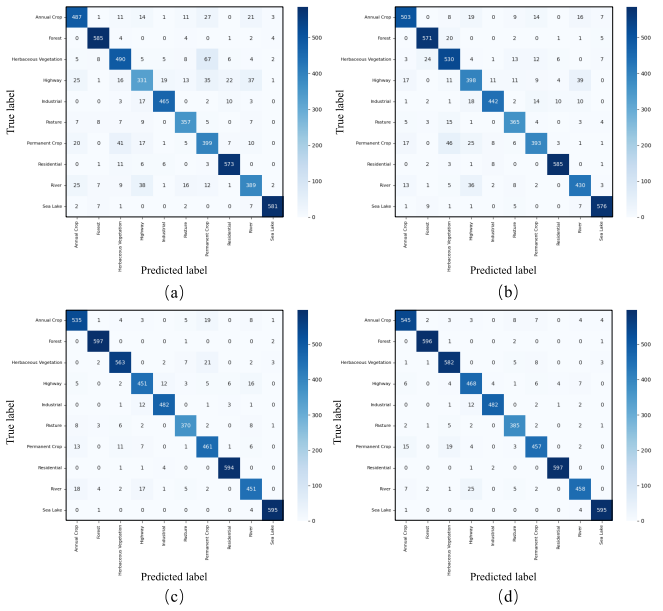


Fig. 11. Confusion matrices on the EuroSAT validation set. (a) Linear probing confusion matrix of MoCo-v2. (b) Linear probing confusion matrix of SwiMDiff. (c) Fine-tuning confusion matrix of MoCo-v2. (d) Fine-tuning confusion matrix of SwiMDiff.

the boundaries distinguishing "Highway" from "River" and "Industrial" are clearer. The confusion matrices in Fig. 11 present specific classification results, also demonstrating the advanced nature and superiority of our proposed SwiMDiff.

b) Ablation results: Table V also displays the classification ablation results of each module on the EuroSAT dataset. For linear probing, MoCo-v2 + SwiM and MoCo-v2 + Diff, respectively, improve by 1.9% and 1.7% over the baseline, while for fine-tuning, they improve by 0.9% and 0.6%, respectively. These results indicate the effectiveness of these modules in extracting discriminative information and semantic details, underscoring our method's versatility and potential applicability to various remote sensing datasets.

V. Conclusion and Future Work

To enhance the efficacy of CL in remote sensing, we introduce SwiMDiff, a novel self-supervised pretraining framework. Initially, our approach incorporates a scene-wide matching strategy into CL. This strategy introduces intraclass similarity by treating images from the same scene as false negatives, thereby effectively addressing sample confusion and enhancing representation learning. In addition, SwiMDiff integrates the diffusion model with CL. It utilizes pixel-level diffusion constraints to amplify the encoder's ability in detail extraction, particularly emphasizing fine-grained details in RSI. These two aspects complement each other, collectively strengthening the encoder's ability to extract both global and local features from RSI. SwiMDiff contributes a richer and more transferable representation for remote sensing, presenting a new self-supervised solution for unlabeled RSIs.

Under consistent conditions with regard to pretraining datasets and backbone networks, we compare SwiMDiff with state-of-the-art self-supervised methods. Experimental results demonstrate the superiority of SwiMDiff in change detection on OSCD and LEVIR-CD, as well as land-cover classification on BigEarthNet and EuroSAT.

In the continuation of our work, we will focus on compressing and accelerating the network framework integrated with the diffusion model, aiming to reduce the computational resources required for image self-supervised representation learning to make it more practical and efficient.

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